MIL-A-43146B
14 May 1985
SUPERSEDING
MIL-A-43146A
9 March 1971

MILITARY SPECIFICATION

ACCESSORY SET, PARACHUTE, SCUBA ADAPTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers an accessory set consisting of the following components:

- Support straps
- SCUBA tank protector shield
- Back strap
- Waistband extension strap

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Fibrous Cotton, Wrapping
PPP-B-26	-	Bag, Plastic, (General Purpose)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-I-6903	-	Ink, Marking (For Parachutes and Other Textile Items)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-147	-	Palletized Unit Loads

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

11-1-308	-	Accessory Set, Parachute, SCUBA Adapting; Set, Complete

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials. Materials when specified shall conform to the applicable documents and requirements specified herein.

3.2.1 Marking medium. The marking medium shall be ink, color blue, No. 15102, conforming to MIL-L-6903.

3.2.2 Color. The webbing, tape, cloth, and the thread shall be dyed Olive Drab, and shall show fastness to weathering equal to or better than the standard sample. If no standard sample is available,the color shall show good fastness to weathering.

3.3 Construction. The construction shall conform in all respects to No. Drawing 11-1-308 and as specified herein.

3.3.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn. The lock shall be imbedded in the materials sewn.

3.3.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the packboard strap shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

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3.3.1.2 Automatic stitching. Automatic stitching may be used to perform the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.3.1.3 Thread ends. All thread ends shall be trimmed to a length of not more than 1/4 inch.

3.4 Marking. Each component shall be marked for identification with the item name, part number, manufacturers name or identifiable trade mark, and year of manufacture conforming to MIL-STD-130.

3.4.1 Straps. All straps shall be marked with 1/2 inch characters on the free end of each strap. Marking medium shall be as specified in 3.2.1.

3.4.2 Shield, SCUBA tank protector. The markings shall be permanently indented with characters of such size as to be clearly legible. The markings shall be placed on the outer surface and centered at the center line location shown on the drawing.

3.5 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

3.6 Workmanship. The components of the accessory set shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2. In-process inspection. Inspection of the components and subassemblies shall be made at appropriate stages of fabrication to determine conformance to the requirements of this document and the applicable drawing. The Government reserves the right to exclude from consideration for acceptance components for which in-process inspection has indicated non-conformance.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of sets. The sample unit shall be one set. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Design	Any characteristic not in accordance with the specified requirements	X	
Assembly of accessory set	Incomplete set	X	
Construction and workmanship	Any component omitted loose, or otherwise not securely retained	X	
Cleanness	Dirt, grease, oil, or foreign matter		X
Stitches and stitching	Not type of stitch specified	X	
	Stitch pattern not as specified	X	
	One stitch less than minimum specified		X
	Two or more stitches less than minimum specified	X	
	One or more stitches in excess of maximum specified		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Stitches and stitching (cont'd)			
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling fabric to sew over heavy places, heavy seams, or in turning corner shall be classified as follows: (a) Within the Major defect classification - Minor defect (b) Within the Minor defect classification - No defect		
	Thread breaks and skipped stitches not overstitched a minimum of 2 inch past ends of stitching defect		X
	Loose tension, resulting in loosely exposed top or bobbin thread; or tight tension, resulting in tightness or puckering of material at stitch line		X
	Thread ends not trimmed		X
	Any row of stitching missing	X	
	Needle chews	X	
	Stitching pattern not complete	X	
	Stitching run off at edge and not restitched in accordance with specified stitch pattern		X
	Stitching ends backstitched less than 1/2-inch (unless secured in a hem or by other stitching)		X
Nylon webbing or duck (as applicable)	Ends not seared where required		X
	Cut, tear, hole, or abrasion marks	X	
	Thin or weak place	X	
	Edges frayed, cut, or torn	X	
	Edge beaded or corded	X	
	Color not as specified		X
	Any splices (not continuous)	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Hardware and shield (as applicable)	Fractured, malformed, or bent	X	
	Sharp burr or metal sliver	X	
	Finish peeling, flaked, blistered, or otherwise nonadherent	X	
	Any area of rust or corrosion or area of no finish	X	
Identification marking	Missing, illegible, incomplete, or incorrect		X
	Improper location or not applied as specified		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimension specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of sets. The sample unit shall be one set. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE II. Packaging defects

Examine	Defect
Marking (interior and exterior container)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing, damaged, defective, or not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping or taping, inadequate stapling. Bulged or distorted container.
Contents (exterior and interior container)	Number per container not as specified.

4.4.6 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. Palletization defects

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Each strap shall be compactly folded and securely tied with twine conforming to type I or II, 8-ply of T-T-871. The four folded and tied straps required for one accessory set shall be unit packed in a close-fitting, 0.0002-inch thick polyethylene bag conforming to type I or II, style 1 or 2 of PPP-B-26. Prior to or during the closure operation, excess air within the bag shall be expelled. The bagged straps shall be securely tied to the inside surface at the center line of the protector shield with the specified twine.

5.1.1.2 Intermediate packing. Two accessory sets, unit packed as specified in 5.1.1.1, shall be packed in the most compact manner in a snug-fitting intermediate box conforming to style RSC (end-opening), grade W6c of PPP-B-636. The box closed and waterproofed with tape in accordance with the appendix of PPP-B-636.

5.1.2 Commercial. Accessory sets shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twelve accessory sets, preserved as specified in 5.1, shall be packed in a snug-fitting cleated plywood shipping container conforming to overseas type, style A or I, grade A type 2 load of PPP-B-601. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Twelve accessory sets, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard. When specified (see 6.2), the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

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5.2.3 Commercial packing. The accessory set preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2) accessory sets, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, intermediate packs shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129, or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information listed below, legibly printed or stamped in black directly on the bag across the center face, or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

ACCESSORY SET (STRAP COMPONENTS)
SUPPORTED STRAPS - 2 EACH
BACK STRAP - 1 EACH
WAISTBAND EXTENSION STRAP - 1 EACH

6. NOTES

6.1 Intended use. The accessory set is intended for use in adapting self contained, underwater breathing apparatus (SCUBA) to the Parachute, Personnel, Troop, A/P28S-11, Air Force Part No. 62D4316.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document
b. First article (see 3.1, 4.3, and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When palletization is required (see 5.3).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples of webbing, tape, cloth, and thread, address the contracting officer issuing the invitation for bids.

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6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of one complete set of the components listed in 1.1. The contracting officer should include specific instructions in all acquisition documents regarding arangements for inspection and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL
Air Force - 99
Navy - SH

Review activities:

Army - AV
Air Force - 82

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-43146B	ACCESSORY SET, PARACHUTE, SCUBA ADAPTING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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